Vector space and Subspace

1 Different sets, common properties

Let us look at the following examples and note the common properties.

Example 1. Consider the set $V = \mathbb{R}^3 = \{(x_1, x_2, x_3) : x_i \in \mathbb{R}\}$. For any two elements

$$x = (x_1, x_2, x_3), \quad y = (y_1, y_2, y_3)$$

define the addition and scaling operations as

$$\begin{aligned} x + y &= (x_1 + y_1, x_2 + y_2, x_3 + y_3) \\ \alpha \cdot x &= (\alpha \cdot x_1, \alpha \cdot x_2, \alpha \cdot x_3), \quad \alpha \in \mathbb{R} \end{aligned}$$

We note that $x + y$ and $\alpha \cdot x$ are again in the set $V = \mathbb{R}^3$.

Example 2. Let $\mathcal{P}_{100}$ denote the set of all polynomials in the variable x, with real numbers as coefficients and of degree at most 100. That is

$$\mathcal{P} = \{a_0 + a_1 x + \cdots + a_{100} x^{100} : a_i \in \mathbb{R}\}$$

For any two polynomials of degree $p(x)$ and $q(x)$ in $\mathcal{P}_{100}$, define the addition and scaling operations as

$$\begin{aligned} p(x) + q(x) &= (a_0 + a_1 x + \cdots + a_{100} x^{100}) + (b_0 + b_1 x + \cdots + b_{100} x^{100}) \\ &= (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_{100} + b_{100}) x^{100} \\ \alpha \cdot p(x) &= \alpha \cdot (a_0 + a_1 x + \cdots + a_m x^m) = \alpha \cdot a_0 + \alpha \cdot a_1 x + \cdots + \alpha \cdot a_m x^m \end{aligned}$$

We note that $p(x) + q(x)$ and $\alpha \cdot p(x)$ belong to the set $\mathcal{P}$.

Example 3. Let $M_{2\times 3}(\mathbb{R})$ be the set of all 2×3 matrices with real entries. For any two matrices a and b of this set define the addition and scaling operations as

$$a + b = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \end{pmatrix}$$

$$\alpha \cdot a = \alpha \cdot \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} = \begin{pmatrix} \alpha \cdot a_{11} & \alpha \cdot a_{12} & \alpha \cdot a_{13} \\ \alpha \cdot a_{21} & \alpha \cdot a_{22} & \alpha \cdot a_{23} \end{pmatrix}$$

We note that $a + b$ and $\alpha \cdot a$ belong to the set $M_{2\times 3}(\mathbb{R})$.

What are the common observations in the above examples?

- We have used two operations: addition and scaling, but we need to define them depending on the set we are working with.
- Given any two elements of the set, their sum as well as the scalar multiples of these elements belong to the set.

In mathematics, if many examples are seen that have similar properties, these are unified as a theory. The advantage is that a general result or property is studied and the examples become particular cases.

In the present case, the name of the unified theory is Linear Algebra and the space in which we carry out the operations of addition and scaling is called Vector space.

The fundamental property of a vector space is that whenever the elements of a vector space interact among themselves via the operations of addition and scaling, we get an element that is still inside the space.

2 Vector Spaces

The formal definition of a vector space is the following.

Definition of a vector space

Definition 1. Consider a non-empty set V and define the operations $+ : V \times V \to V$ called addition and $* : \mathbb{R} \times V \to V$ called scaling satisfying the following properties

- $u + v = v + x$, (commutativity of addition)
- $(u + v) + w = u + (v + w)$ (associativity of addition)
- There exists $\Theta \in V$ such that $u + \Theta = u$ (existence of additive identity)
- Given any $v \in V$ there exists $u \in V$ such that $u + v = \Theta$, (existence of additive inverse). We denote the additive inverse of u by $-u$.
- (*Note that the scalar α belongs to the set of real numbers, on which we also have addition and multiplication.*) The following are the compatibility relations among the addition and multiplication of real numbers and the addition and scaling that we have defined on V.
  - $\alpha * (u + v) = \alpha * u + \alpha * v$, $(\alpha + \beta) * u = \alpha * u + \beta * u$
  - $(\alpha\beta) * u = \alpha * (\beta * u)$
- $1 * u = u$, (unitary scaling)

Under these operations $+$ and $*$, the set V is called a **vector space** over $\mathbb{R}$ (or a real vector space) and elements of V are called **vectors**.

Example 4. Show that $\mathbb{R}^n$ is a vector space with the usual vector addition and scaling.

For $u = (u_1, u_2, \cdots, u_n)$, $v = (v_1, v_2, \cdots, v_n)$ and $w = (w_1, w_2, \cdots, w_n)$ in the set $\mathbb{R}^n$, commutativity and associativity of addition holds. The additive identity is the n-tuple $\Theta = (0, 0, \cdots, 0)$. The additive inverse of u is $-u = (-u_1, -u_2, \cdots, -u_n)$. For the compatibility relations we have

$$\begin{aligned}
\alpha * (u + v) &= \alpha * (u_1 + v_1, \cdots, u_n + v_n) = (\alpha * (u_1 + v_1), \cdots, \alpha * (u_n + v_n)) \\
&= (\alpha * u_1, \cdots, \alpha * u_n) + (\alpha * v_1, \cdots, \alpha * v_n) = \alpha * u + \alpha * v \\
(\alpha + \beta) * u &= (\alpha + \beta) * (u_1, \cdots, u_n) = ((\alpha + \beta) * u_1, \cdots, (\alpha + \beta) * u_n) \\
&= (\alpha * u_1, \cdots, \alpha * u_n) + (\beta * u_1, \cdots, \beta * u_n) = \alpha * u + \beta * u
\end{aligned}$$

The unitary scaling property holds as

$$1 * u = 1 * (u_1, \cdots, u_n) = (1 * u_1, \cdots, 1 * u_n) = (u_1, \cdots, u_n) = u$$

The spaces $\mathbb{R}^n$ are fundamental examples of vector spaces. Geometrically, elements of these spaces are called vectors and are denoted by u, v, w. The name stays when we discuss other vector spaces. In examples 2 and 3 earlier, $\mathcal{P}_{100}$ and $M_{2\times 3}(R)$ are also vector spaces (*prove!*) and $u = x^3 + 2x + 1$, $v = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, for example, are vectors in their respective vector spaces.

Example 5. The sets $\mathcal{P}_{100}$ and $M_{2\times 3}(R)$ are also vector spaces. The additive identity for $\mathcal{P}_{100}$ is $\Theta = 0 + 0x + 0x^2 + \cdots + 0x^{100}$ and for $M_{2\times 3}(R)$ is $\Theta = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Think why these are the additive inverses in the respective vector spaces. Can you find the respective additive inverses? Also, verify the other properties.

Note 1. As you might have observed, the additive identity is always denoted by Θ and its exact expression depends on the vector space we are in. Also, for ease of notation, we write the scaling property $\alpha * u$ as αu.

Let us look at examples that verifies the idea that additive identity and additive inverse always depend on how we define the operations of addition and scaling. These examples also show how a set is a vector space under some definition of $+$ and $*$ but is not a vector space under some other definition.

Example 6. Show that the space $\mathbb{R}^2$ with the operations $(x_1, y_1) + (x_2, y_2) = (x_1 + y_2, y_1 + x_2)$ and $\alpha * (x, y) = (\alpha * x, \alpha * y)$ is not a vector space.

Let $u = (x_1, y_1)$ and $v = (x_2, y_2)$. Then $u + v = (x_1, y_1) + (x_2, y_2) = (x_1 + y_2, y_1 + x_2)$ while $v + u = (x_2, y_2) + (x_1, y_1) = (x_2 + y_1, y_2 + x_1)$. Hence commutativity of addition fails. Does associativity holds?

Let us verify if the additive identity exists. Suppose $\Theta = (a, b)$ is such that $u + \Theta = u$ and $\Theta + u = u$. This means $(x_1 + b, y_1 + a) = (x_1, y_1)$ and $(a + y_1, b + x_1) = (x_1, y_1)$. The first equality gives $b = 0, a = 0$ while the second equality gives $a = x_1 - y_1$, $b = y_1 - x_1$. Thus additive identity does not exist. (It will exist *only* in the case $x_1 = y_1$ but by definition we require the additive identity property for all vectors.)

Do the compatibility relations hold in this case?

Example 7. Consider the space $\mathbb{R}^2$ with the operations

$$(x_1, y_1) + (x_2, y_2) = (x_1 x_2, y_1 y_2) \text{ and } \alpha * (x, y) = (\alpha * x, \alpha * y)$$

Verify the commutative and associative property. To determine the additive identity $\Theta = (a, b)$, by definition we have

$$(x_1, y_1) + \Theta = (x_1, y_1) + (a, b) = (x_1 a, y_1 b) = (x_1, y_1)$$
$$\Theta + (x_1, y_1) = (a, b) + (x_1, y_1) = (a x_1, b y_1) = (x_1, y_1)$$

Both the cases give $a = 1, b = 1$. Hence the additive identify under the given definition of addition is $(1, 1)$. Let (c, d) be the additive inverse of any element (x_1, y_1). Then

$$(x_1, y_1) + (c, d) = \Theta \Rightarrow (x_1 c, y_1 d) = (1, 1) \Rightarrow c = 1/x_1, d = 1/y_1.$$

Hence the additive inverse for any element (x, y) under the given definition of addition is $(1/x, 1/y)$. However, the additive inverse exist only if $x \neq 0$ and $y \neq 0$. Thus non-zero vectors like $(8, 0)$ or $(0, 1/2)$ do not have an additive inverse. Thus under the given definition of addition $\mathbb{R}^2$ is not a vector space.

Example 8. Consider the space $\mathbb{R}^2$ with the operations

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \text{ and } \alpha * (x, y) = (x^\alpha, y^\alpha)$$

The properties of commutativity, associativity, existence of additive identity and additive inverses are satisfied because of the way addition is defined. Verify the compatibility relations for the given definition of scaling.

To consolidate, the properties of commutativity, associativity, existence of additive identity and additive inverses, and the compatibility properties depend on how one chooses to define the operation of addition and scaling. In some cases, the set becomes a vector space while in other cases, it does not.

3 Subspaces of a vector space

A subspace is a subset of the vector space which inherits the same structure as that of the vector space. Remember, a vector space is a set and can have many subsets. But only those subsets that satisfy the following definition are subspaces of the given vector space.

Let V be a vector space on which the operations of addition '$+$' and scaling '$*$' have been defined. The following gives us the formal definition of a subspace.

Definition of a subspace

Definition 2. Let W be a non-empty subset of a vector space V. Then W is said to be a subspace of V if

- The additive identity Θ of V also belongs to W.
- If the vectors $w_1, w_2 \in W$, then $w_1 + w_2 \in W$.
- If $\alpha \in \mathbb{R}$ and $w \in W$, then $\alpha w \in W$.

The main idea of a subspace is that just like the vector space, when we add or scale elements of a subspace, we get an element that still belongs to the subspace.

Example 9. Consider the vector space of real numbers $\mathbb{R}$ (verify!) and the subsets $\mathbb{Q}$ of rational numbers and $\mathbb{R} \setminus \mathbb{Q}$ of irrational numbers.

- If we add any two rational numbers, we still get a rational number and hence an element of the subset $\mathbb{Q}$. But if we take $\alpha = \sqrt{2}$, then the scale $\alpha * p = \sqrt{2}p$ doesnot belong to $\mathbb{Q}$ for any rational number p. Hence the set of rational numbers is not a subspace of the real numbers.
- If we add any two irrational numbers, it is not necessary that we always get an irrational number, for example $2 + \sqrt{2}$ and $-\sqrt{2}$. Thus the set of irrational numbers is also not a subspace of the real numbers.

Example 10. Consider the set $\mathcal{P}_3 = \{a_0 + a_1x + a_2x^2 + a_3x^3 : a_i \in \mathbb{R}\}$ consisting of all polynomials of degree at most 3. It is a subset of $\mathcal{P}$ and is also a vector space with the operations defined on $\mathcal{P}$.

Consider the vectors $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ and $q(x) = b_0 + b_1x + b_2x^2 + b_3x^3$. Then $p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + (a_3 + b_3)x^3 = q(x) + p(x)$. The associativity of addition also follows. The additive identity is $\Theta = 0 + 0x + 0x^2 + 0x^3$ and the additive inverse is $-p(x) = -a_0 - a_1x - a_2x^2 - a_3x^3$. The other properties also follow.

Let us look at some subsets of a parent space that are not subspaces.

Example 11. Consider the following subsets of the parent space $\mathbb{R}^2$

$$W_1 = \{(x_1, x_2) : x_1x_2 \geq 0\}, \quad W_2 = \{(x, x) : y = x^2\}, \quad W_3 = \{(x, y) : y = x + 1\}$$

The condition $x_1x_2 \geq 0$ implies that x_1 and x_2 have the same sign. That is (x_1, x_2) lies in the first or the third quadrant along with the co-ordinate axes. This is clearly a subset of $\mathbb{R}^2$. The vectors $(0, 1)$ and $(-1, 0)$ belong to W_1, but their sum $(-1, 1)$ does not belong to W_1.

Geometrically, W_2 represents the parabola $y = x^2$. Take any point $(2, 4)$ in W_2. Its scaling $2(2, 4) = (4, 8)$ does not belong to W_2.

Similarly, W_3 represents the straight line $y = x + 1$ which does not pass through origin. The point $(0, 0)$ which is the additive identity in $\mathbb{R}^2$ does not belong to W_3.

What are the reasons for which a subset W of a parent space V might not become a subspace?

- If $w_1, w_2 \in W$, then w_1, w_2 belong to V. Since V is the parent space, $w_1 + w_2$ and αw_1 belong to V. But it is not necessary that $w_1 + w_2$ and αw_1 belong to W.
- The additive identity Θ of V need not belong to W.

We also note that many properties of V hold in W.

- Viewing $w_1, w_2 \in W$ as vectors in the vector space V, they satisfy the commutativity and associativity properties of addition.
- The compatibility relations with any real scalars α, β and the property of unitary scaling also hold. Again the reason is that $w_1, w_2 \in W$ are also vectors in V.

Let us look at some examples of subsets of vector spaces and how to verify if they are subspaces.

Example 12. Consider the subset $W = \{(x_1, x_2) : x_1x_2 = 0\}$ of $\mathbb{R}^2$. The condition $x_1x_2 = 0$ implies that $x_1 = 0$ or $x_2 = 0$. Geometrically, this is the subset of the points lying on the co-ordinates axes in $\mathbb{R}^2$. It is clear that $\Theta = (0, 0) \in W$. But consider $w_1 = (1, 0)$ and $w_2 = (0, 1)$. Then $w_1, w_2 \in W$ but $w_1 + w_2 = (1, 1)$ does not belong to W. Hence W is not a subspace of $\mathbb{R}^2$.

Example 13. Consider the subset $W = \{(x_1, y_1) : x_1, y_1 \in \mathbb{Z}\}$ of $\mathbb{R}^2$. The condition implies that both the coordinates of the pair (x_1, y_1) should be integers. Again $\Theta = (0, 0) \in W$. If $w_1 = (x_1, y_1)$ and $w_2 = (x_2, y_2) \in W$ then $w_1 + w_2 = (x_1 + x_2, y_1 + y_2) \in W$ as $x_i + y_i$ is also an integer if x_i and y_i are integers. However, the scaling property is not satisfied as for $\alpha = \sqrt{2}$, $\alpha(1, 2) = (\sqrt{2}, 2\sqrt{2})$ does not belong to W. Hence W is not a subspace of $\mathbb{R}^2$.

Example 14. Consider the subset $W = \{(x, y) : 2x - 5y = 0\}$ of $\mathbb{R}^2$. The condition shows that W is the line $y = 2x/5$ passing through origin in $\mathbb{R}^2$. Then $\Theta = (0, 0) \in W$. If $w_1 = (x_1, y_1)$ and $w_2 = (x_2, y_2)$ are in W, then $w_1 + w_2 = (x_1 + x_2, y_1 + y_2)$ also belong to W as $2(x_1 + x_2) - 5(y_1 + y_2) = 2x_1 - 5y_1 + 2x_2 - 5y_2 = 0 + 0 = 0$. Again, $\alpha w_1 \in W$ as $\alpha w_1 = (\alpha x_1, \alpha y_1)$ and $2(\alpha x_1) - 5(\alpha y_1) = \alpha(2x_1 - 5y_1) = 0$. Hence W is a subspace of $\mathbb{R}^2$. Note that W is the solution space of the homogeneous system of equation $2x - 5y = 0$.

Example 15. Consider the subset W of 2×2 symmetric matrices of the vector space $M_{2\times2}(\mathbb{R})$ of 2×2 matrices with real entries. In this case Θ is the 2×2 zero matrix and it belongs to W since it is also symmetric. The sum of two symmetric matrices is also symmetric and any scaling of a symmetric matrix is also symmetric. Hence W is a subspace.

Example 16. The trace of an $n \times n$ matrix A, denoted $tr(A)$ is the sum of diagonal entries of A. The set $N_{n\times n}(\mathbb{R})$ of $n \times n$ matrices having trace equal to 0 is a subspace of the set $M_{n\times n}(\mathbb{R})$ of all $n \times n$ matrices with real entries. Here Θ is the $n \times n$ zero matrix. If $A, B \in N_{n\times n}(\mathbb{R})$, then $A + B \in N_{n\times n}(\mathbb{R})$ since $tr(A + B) = tr(A) + tr(B) = 0$. Again $\alpha A \in N_{n\times n}(\mathbb{R})$ as $tr(\alpha A) = \alpha tr(A) = 0$.

Exercises

1. Let $W = \{(a, a + 1) : a \in \mathbb{R}\}$ be a subset of the vector space $\mathbb{R}^2$ with the standard definitions of addition and scaling. Determine whether W is a subspace of $\mathbb{R}^2$. (NO)

2. In the following exercises determine whether the subset S is a subspace of $\mathbb{R}^3$.

   (a) $S = \{(x_1, x_2, x_3)|x_1 + x_3 = 2\}$, (NO)

   (b) $S = \{(x_1, x_2, x_3)|x_1x_2x_3 = 0\}$, (NO)

   (c) $S = \{(s - 2t, s, s + t)|s, t \in \mathbb{R}\}$, (YES)

   (d) $S = \{(x_1, a, x_3)|x_1, a, x_3 \in \mathbb{R}\}$, (YES)

3. In the following exercises determine whether the subset S is a subspace of $\mathcal{P}_5$.

   (a) S is the set of all polynomials with exact degree 5. (NO)

   (b) S is the set of all polynomials with even degree. (YES)

   (c) S is the set of all polynomials of the form $p(x) = ax^2$. (YES)

   (d) S is the set of all polynomials of degree less that or equal to 4. (YES)